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HOLE	SAMPLE No.	FOOTAGE	COLOUR	AGGREGATE.	MINERALS			COPPER MINERALS	RATIO Sulph: O. Salts	ASSAY % Cu.	REMARK
L 1	1347	0' - 5'	Orange White	Sand chips of L. SStone Rounded grains of White Qtz Clay	Qtz, Kaolin Gypsum, Clay.			Not Noted			L. SStone
	1348	5' - 10'	Off White	Chips of L. Sst α grains well rounded	As above			Not Noted			L. SStone
	1349	10' - 15'	Off White	As above	As above			Not Noted			L. SStone
	1350	15' - 20'	Yellowish	Well rounded grains of Qtz	— " —			— " —			L. SStone
	1351	20' - 25'	Off White	— " —	— " —			— " —			L. SStone
	1352	25' - 30'	Off White	— " —	— " —			— " —			L. SStone
	1353	30' - 35'	Brownish	As above + Chips of R Qtzite α Fe Stained Clay.	— " —			— " —			L. SStone & R. Qtzite
	1354	35' - 40'	Pink.	Angular chips of R Qtz. minor clay	Qtz. clay.			Malachite α Chalcocite	1 : 4		R Qtzite
	1355	40' - 45'	Pink.	— " —	— " —			— " —	4 : 1		R. Qtzite
	1356	45' - 50'	— " —	— " —	— " —			— " —	1 : 1		R. Qtzite
	1357	50' - 55'	— " —	— " —	— " —			— " —	4 : 1		R. Qtzite
	1358	55' - 60'	"	"	"			"	4 : 1		R. Qtzite
	1359	60' - 65'	"	"	"			"	4 : 1		"
	1360	65' - 70'	Redish	"	"			Few Specs noted	4 : 1		"
	1361	70' - 75'	Redish	"	"			— " —	— " —		R Qtzite. R Qtzite
	1362	75' - 80'	"								
	1363	80' - 85'	"					Not Noted.	— " —		R Qtzite
	1364	85' - 90'	"					"	"		"

Cont

HOLE	SAMPLE No.	FOOTAGE	COLOUR	AGGREGATE.				COPPER MINERALS	RATIO Sulph : O. Salts	ASSAY % Cu.	REMARKS
L1	1365	90-95'	Pink	Angular chips				Not Noted			R. & Tzite
	1366	95-100'	"	of Quartz.				"			"
	1367	100-105'	"	Some Clay				"			"
	1368	105-110'	"	"				"			"
	1369	110-115'	"	"				"			"
	1370	115-120'	"	"				"			R. & Tzite
	1371	120-125'	"								
	1372	125-130'	"								
	1373	130-135'	"								
	1374	135-140'	"								
	1375	140-145'	"								
	1376	145-150'	"								
	1377	150-155'	"								
	1378	155-160'	"								
	1379	160-170'	"								
	13	170-180'	"	As Above				As Above			R. & Tzite

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PACMINEX PTY LIMITED

PETROGRAPHY OF MOUNT GUNSON SEDIMENTS

PART I SURFACE CHARACTERISTICS OF QUARTZ GRAINS

PART II A PETROGRAPHIC STUDY BY THIN-SECTION

SYDNEY

October, 1972.

P.J. CURTIS

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INDEXPAGE NO.PART I SURFACE CHARACTERISTICS OF QUARTZ GRAINS

1.	INTRODUCTION	1
2.	SUMMARY	2
3.	DISCUSSION	3
4.	MICROSCOPY	6
5.	REFERENCES	

PART II A PETROGRAPHIC STUDY BY THIN-SECTION

1.	INTRODUCTION	13
2.	SUMMARY OF OBSERVATIONS	14
3.	DISCUSSION SUMMARY	22
4.	DESCRIPTION OF SECTIONS	36
5.	REFERENCES	75

000067

PART I

Surface characteristics of quartz grains from the
Pandurra Quartzite and Whyalla Sandstone, Mt. Gunson Prospect.

000068

1. INTRODUCTION

During June 1972 the writer undertook a field investigation for 8 days at Mt. Gunson. Part of the time was spent sampling percussion drill samples of Pandurra Quartzite and Whyalla Sandstone from various parts of the mineral leases. The samples were dry screened to + 10 mesh and the + 10 mesh fractions were examined under a low power stereomicroscope to study surface characteristics of the quartz mineral grains.

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2. SUMMARY

2.1 The Pandurra Quartzite and Whyalla Sandstone are each remarkably uniform over wide areas.

2.2 The Pandurra Quartzite is mainly a poorly sorted but fine-grained quartzite. Thin individual grains are quite non-uniform, irregular and angular or at the best, subangular. The grains have clear surfaces mostly but occasionally are lightly frosted and frequently are tinted pink by adhering hematite particles. The original cement of hematite and clay has been largely replaced by quartz.

2.3 The Whyalla Sandstone is largely of aeolian origin intermixed with lacustrine silt and clay. The grains are in most cases quite round by surface frost, scour and sub-parallel scratching of the grains (particularly the large crystals) indicate an aeolian origin for them. Some intermixing with subangular littoral dune sand is seen and absence of aeolian false bedding and the presence of much clay/hematite matrix indicated removal by flood-water of the dune sand and intermixing with adjoining playa lake mud and re-deposition near the strand line in flood channels.

3. DISCUSSION

Sampling of Pandurra Quartzite and Whyalla Sandstone from percussion drill holes over a large area including areas adjoining the Mill, at Gunyah Lake, the House; and further afield from the Cattle Grid and Cane Grass Dam shows that the two stratigraphic units are quite distinct.

3.1 The Pandurra Quartzite grains show :-

3.1.1 Irregular water-worn grains as channel-fill or estuarine deposits dumped on the edge of a once sharply rising hinterland and as such are poorly sorted;

3.1.2 Any frosting of these grains is slight and is due to attenuated buffeting of the grains in a water medium;

3.1.3 Subsequent diagenesis after sedimentation and compaction has caused the quartz grains to be welded together by silica addition, or by replacement of existing clay and hematite from percolating waters containing silica, the mosaic being interrupted only by a little residual kaolin, hematite or sulphide cement.

3.2 Whyalla Sandstone is distinct in that :-

3.2.1 It is a sandstone containing a great deal of clay/sericite/hematite matrix;

3.2.2 The maximum overall grain size is comparatively large and the grains are frequently perfectly round, show heavy frosting, scour and sub-parallel scratches which are most likely the result of aeolian influences since the degree and

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type of attrition is similar to that observed in desert dune sands;

3.2.3 However, the large amount of fine matrix present suggests secondary deposition of this dune sand by flood-waters to include additional fine material from adjoining playa lakes;

3.2.4 Sizing of the quartz grains at the transfer stage was already completed by the previous aeolian activity and maintenance of a more or less graded sequence shows the short distance travelled to redeposition;

3.2.5 Resettling in water, is indicated by the presence of large clasts and absence of aeolian false bedding (Ref. 1).

Quartz grains with a secondary growth of silica in optical continuity with the detrital nucleus (Ref. 2) is characteristic of the Pandurra Quartzite and occurs best in open sands with no interstitial cement when the secondary silica assumes a remarkably good crystalline form (Ref. 3) In this situation it probably shows a prolonged deposition of sand grains under water which is not very deep since the deposition of secondary silica indicated proximate alkaline waters changing to slightly acid or neutral pH waters in the zone of deposition. Taking into account the poor sorting of the grains and the hypothesis of a sharply rising hinterland to explain this, a high proportion of fine silt and clay matrix would be expected but has seemingly been washed out by strong current action or was removed during the subsequent diagenetic phase.

The presence of subangular as well as of rounded frosted grains in the Whyalla Sandstone may show some exchange of aeolian and littoral grains by coastal dunes by wind action

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with those from desert dunes more remote from the strand (Refs. 2 and 5).

Pettijohn (Ref. 4) suggests that the frosting of sand grains is due to aeolian action and that a sand of mixed parentage will contain grains with a diversity of textural features and that the last epoch of transportation and deposition was inadequate to erase the surface markings of previous cycles. Thus the elimination of frost from some of the Whyalla grains and elongation of some grains with some gouging may indicate an overprint of aeolian conditions onto characteristics of severe water-worn activity. A fine vitreous glaze or polish (Ref. 4) formed by fine particles sand-blasting quartz grains is seen in the Whyalla Sandstone but is not common possibly because when present it was removed by subsequent diagenesis after deposition of the grains in water.

According to Margolis and Krinsley (Ref. 5) who collected desert and coastal dune sands of aeolian origin from 20 localities around the world, chemical solution and re-deposition of silica by the action of desert dew along with mechanical wind abrasion are factors contributing towards frosting and rounding of sand grains. The deep-pitting of the Whyalla grains associated with the frosting which frequently predominates at projecting corners or edges if the grain has a tendency to be subangular, suggests dominant wind induced grain impacts as the agent responsible for the shape and appearance of these grains.

Frosting and rounding in aeolian attrition is generally more intense with the larger grains according to Kuenen quoted by Margolis and Krinsley (Ref. 5) and Kuenen also stated that rounding is primarily of mechanical aeolian origin. Margolis and Krinsley demonstrated experimentally that diagenesis may defrost sand grains.

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Where the dune sand grains are iron stained as per present-day sand dunes at Mt. Gunson it occurs during long periods of reduced wind activity as significant abrasion would remove it from the surface of the grains. Iron staining of the Whyalla Sandstone grains is quite uncommon due to the steady attrition which occurred by continuous wind activity prior to subsequent deposition in shallow water.

4. MICROSCOPY

4.1 Pandurra Quartzite

4.1.1 House Workings

Welded pink grains mainly clear but some larger grains are lightly frosted. The individual grains average between 0.2 and 1.0 mm (mainly 0.5 mm) and are poorly sorted, mainly angular though some rounded grains occur and grains do vary from quite angular to rounded in shape. The cement is generally quartz, thus fractures occur through grains and matrix in rock fragments rather than through soft matrix as in the Whyalla Sandstone. However not all the cementing material is quartz and where kaolinite > hematite/limonite > sericite are intergranular cements, euhedral forms as pyramids or prisms can be seen as overgrowths on otherwise angular grains.

4.1.2 Gunyah Lake

Similar to the Pandurra Quartzite examined from the House Workings but the rock cement (apart from quartz) is largely hematite. The quartz grains are irregular in shape and size, poorly sorted, with mainly clear surfaces but with some slight frosting of grain surfaces locally.

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4.1.3 Mount Gunson Traverse

As for the 'House Workings'; poorly sorted grains; variation in shape with many angular grains and wide variation in size from 0.1 to 1 mm. Grain cement (apart from quartz) is largely hematite with some kaolin.

4.1.4 Cane Grass Dam

The grains are more frequently faintly frosted on their surfaces, are irregular and angular, average 0.4 mm and vary between 0.2 and 1 mm. Apart from quartz, sericite is the common grain cement. Some euhedral overgrowths are seen on grains which line fissures or cavities in the quartzite. One specimen (66327) tends to have a lower average grain size (0.2 mm), the grains being rounded to sub-angular and are well sorted. The grains also have clear surfaces.

4.1.5 Cattle Grid

There is gross contamination by Whyalla Sandstone in these samples, but most grains are angular and of variable size (0.1 to several mm's). The grains are mainly clear with minimal matrix (cement), mainly of quartz or sericite (some hematite and sulphide). Some of the larger grains show faintly frosted surfaces.

4.2 Whyalla Sandstone

4.2.1 House Workings

The matrix is rich in kaolinite and some limonite. The quartz grains are colourless, and mainly rounded with minor subangular/rounded grains with surfaces pitted and uniformly frosted, although some of the smaller grains tend to have some clear

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surface areas. Grain size tends to be somewhat larger than in the Pandurra Quartzite varying in one sample between 0.5 to 3 mms and in another between 0.2 and 1 mm. Overgrowths on worn surfaces of euhedral forms is seen occasionally but is uncommon. Limonite constitutes the matrix of a cherty horizon containing some possible gypsum and found only once during the whole investigation.

4.2.2 Gunyah Lake

Matrix again is heavy with kaolinite and some sericite. The grains in this deposit tend to be large (2 to several mm's), are well frosted and round to subangular/round; smaller grains again are particularly clear at surface. Large 'impact' pits up to 0.1 mm are seen on the large grains and occasionally have 'amoeboid-like' protrusions which show frequently clear surfaces, (secondary, euhedral due to diagenesis ?).

One specimen (48887) contains large sub-angular grains, finely frosted with large 'impact' pits and the grains are 2 to 5 mm's in size. These grains show clear surfaces at edges and corners.

4.2.3 Mount Gunson Traverse

Matrix in these samples which is fairly abundant is hematite and kaolinite with a tendency for the hematite to stick to grain surfaces (diagenetic phenomenon). The grain size tends to be somewhat smaller with many being less than 0.4 mm in diameter maximum up to 1 mm. However one sample is predominantly fine grained but ranges up to 3 mms. Large grains irregular, rounded, (>0.5 mm diameter) are uniformly heavily frosted but small grains (>0.5 mms) are subangular to sub-rounded but with finely etched surfaces. Euhedral overgrowths producing a

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clear surface are seen on some smaller grains.

Sorting of the grains is fair.

4.2.4 Cane Grass Dam

Again, abundant matrix of kaolinite with hematite or kaolinite alone. Grain sizes in samples vary; in one sample approximately 0.5 mm and in all others between 1 and 5 mms; rounded, particularly the larger grains and somewhat flat/rounded and with gouged-out undersurfaces in the larger grains. Frosting and pitting of grain surfaces is common and is intense at corners and edges of some sub-angular grains (sample 66252) and this sample also contains rounded chert pebbles (approximately 0.5 mm diameter). A proportion of grains with clear surfaces are seen in sample 66843 and 66313 where the average grain size is smaller (approximately 0.5 mm for sample 66313). Really large pits on surfaces occur in sample 66358 where the average grain size is approximately 3 mms diameter. Sorting is better in the finer grained samples (e.g. 66483). Some euhedral overgrowths are seen on smaller grains.

4.2.5 Cattle Grid

The matrix in these samples is largely abundant sericite and/or kaolinite; in one case (66719) hematite in addition to sericite and kaolinite; and where silicified (67897), it is quartz. The grain size limits are variable from sample to sample at <0.5 mm; 0.2 to 1 mm; 0.5 to 1 mm; and 1 to 5 mm. The larger grains tend to be subangular to rounded, coarsely frosted, pitted, gouged and scratched. Large deep pits on large grains occur at the expense of the finely frosted surface. Angular grains (sample 67933), up to 1.5 mms, show greater frosting at edges and corners of

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the grains. Large rounded cherts are also seen in sample 67933, and all shapes of Whyalla Sandstone quartz grains, from angular to subangular to round, averaging 0.5 mm are seen in this sample. Smaller grains tend to be more angular and less frosted but grains frequently show some pitting and scratches. A varnish is seen on some part-frosted grains (67718, 67934) which is not related to matrix composition. Euhedral overgrowths on quartz grains are seen but are noted only where secondary silicification has taken place (67897). Sorting in all size ranges is fair to good.

4.2.6 Coogee Beach Sand

Two samples were taken, one from the foreshore near the undercliff on the north side of the bay and another from low tide level on the beach, not far from the first sample.

Both samples show angular to subangular sand grains averaging around 0.3 mm diameter. The larger sand grains do show some light frosting and pitting on projecting and level surfaces and similar to the Whyalla Sandstone the frosting is mainly confined to the larger grains. However scouring and scratching of the grains is noticeably absent from the beach sand.

The beach sand grains are generally finer, more angular and less frosted, (the larger grains in particular), than the Whyalla Sandstone grains although some frosting and pitting are seen on beach sand grains.

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4.2.7 Mount Gunson Dune Sand

Two samples were taken, one from a dune situated between the Mill and House Workings (over inferred Woocalla Dolomite); the other from a dune a few hundred yards south-east of the House Workings situated upon inferred Whyalla Sandstone/Pandurra Quartzite. These samples were seen not to be typical desert dune sand in that sizing of the grains is poor, varying between 0.1 and 1 mm with an average size of 0.4 mm. The grains are subangular to rounded as for the Whyalla Sandstone. Sub-parallel scratches occur roughly parallel to the long axis of the grains (where the grains are ovoid in shape), and gouging and pitting of faces of large grains is seen. Frosting of larger grains is more common than for the smaller grains. Many grains have a surface stain of hematite which tends to give them a polished look. Frosting is generally not so heavy as for the Whyalla sand grains. However, the shape, size, presence of an overall 'frost' on the larger grains, scratching and gouging of surfaces would indicate the grains to be derived from the Whyalla Sandstone and it is hoped that dune sands from the Mount Gunson area well away from the Whyalla Sandstone may shortly be sampled and studied.

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PART II

A petrographic study of rocks from the Pandurra
Quartzite and Tent Hill formations of the Mount Gunson area.

1. INTRODUCTION

During the June 1972 visit to Mount Gunson part of the time was spent in sampling diamond drill cores and inspecting and sampling some exposures of Pandurra Quartzite and Whyalla Sandstone. The specimens were thin sectioned and where appropriate polished sections were prepared.

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2. SUMMARY OF OBSERVATIONS

2.1 Pandurra Quartzite

2.1.1 Main Open Pit

- (i) Abundant diagenetic transformation of originally subangular to rounded quartz grains to angular grains by addition of quartz is seen where quartz extends beyond hematite particle trains and fluid inclusions but within the overgrowths; this added quartz is in optical continuity with the nucleus and is deposited from meteoric waters with resultant replacement of original matrix minerals.
- (ii) Large quartz rock fragments as erratics occur and the smaller quartz mineral and rock grains are poorly sorted, yet occasional very rough sorting into coarse and finer bands is seen.
- (iii) Strained and unstrained grains are not structurally orientated in the 'quartzite' fabric and the strained grains with many quartz gneiss fragments seen in the rock are assumed to be derived from a metamorphic source or provenance.
- (iv) Chert grains are not prominent in this stratigraphic unit.
- (v) Kaolinite matrix is seen where not replaced by chert or $\angle$ quartz.
- (vi) Replacement of kaolinite matrix by cherty cement is common.
- (vii) Replacement of atacamite by $\angle$ quartz and vice versa indicates solution and redeposition of copper values during diagenesis.

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2.1.2 West Lagoon

- (i) Very similar texturally to the Main Open Pit.
- (ii) Large quartz rock clasts occur.
- (iii) If the few specimens collected are indicative then deposition of $\angle$ quartz rather than chert (as at the Main Open Pit) to replace soft matrix is common, suggestive of a more 'reducing' and less 'oxidizing' post depositional conditions were operative.

2.1.3 Mystery/Pernatty Lagoon

- (i) A 'greywacke' type with abundant fine-grained matrix - although most of the latter has been replaced by chert.
- (ii) A multi-chambered 'foraminiferal-looking' organism is seen in the 'greywacke-type' sediment.
- (iii) Abundant matrix (primary kaolinite or secondary chert) appears to be typical of the Pandurra Quartzite in the Mystery deposits.
- (iv) A relatively high quartz rock fragments to quartz mineral grains ratio is typical.
- (v) Poor sorting is seen but no 'rough-banding' as seen in the Main Open Pit samples.
- (vi) Ubiquitous quartz gneiss fragments but no metamorphic heavy minerals (kyanite, sillimanite, staurolite, etc).
- (vii) Abundance of playa-lake matrix should point to the presence of some syngenetic/epigenetic sulphides.

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2.1.4 Woocalla Quarry

- (i) Relatively large variation in grain size, poorly sorted.
- (ii) Where possible to see primary grain boundaries the grains are quite round.
- (iii) Cherty matrix is present but primary chert grains are not abundant.
- (iv) There is a lack of soft clay matrix.
- (v) There are a few rock fragments. Poor sorting, lack of heavy minerals, rounding of primary quartz nuclei indicate an abundance of recycled quartz grains.

2.1.5 Mona Lena

Repetitive (from base to top) coarse angular pebbles, sand then silt shows sporadic nature of flash floods which washes clastic material into estuarine channels.

2.2 Woocalla Dolomite

2.2.1 Sweet Nell

The Basal Shale is mineralised with chalcopyrite which has recrystallised with coarse quartz and carbonate in laminae and is therefore secondary. Possibly precipitated originally under reducing conditions with dolomitic mud. Apparently this is not a typical Basal Shale specimen since it is normally quite argillaceous.

2.2.2 Woocalla Quarry

The dolomite is patchy in appearance with manganese wad dendrites which were exsolved from a

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manganiferous dolomite during weathering and recrystallisation of the carbonate. Copper does not readily substitute for calcium or magnesium and is unlikely to occur in the dolomite lattice and in the absence of argillaceous matter or sulphides it is difficult to place.

2.2.3 Mystery/Pernatty Lagoon

- (i) Dr. Barnet of Central Laboratory identified the carbonate sediment as $(\text{Ca}, \text{Mg}, \text{Mn}) (\text{CO}_3)_2$.
- (ii) From the limited number of specimens examined from the Mystery area the dolomite is seen to be very pure with only traces of chert and no argillaceous minerals. So *Collenia* is likely to have grown on dolomite mud banks.
- (iii) *Collenia* forms are marked in section only by difference in dolomite grain sizes.
- (iv) The only definitely transported material are the detrital quartz grains.
- (v) The presence of post depositional baryte indicates definite saline conditions of deposition so the dolomite could be primary.
- (vi) The occurrence of stylolites shows that extensive interstratal solutions existed after deposition of sediment.
- (vii) Copper in the fresh dolomite examined is most likely present as only specks of chalcopyrite.

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2.3 Whyalla Sandstone

2.3.1 Generally, for all areas, the Whyalla Sandstone is distinct in that :-

- (i) Aeolian origin of the larger sand grains is established as large well rounded grains are seen.
- (ii) The associated grains of (i) are a matrix of fine angular to rounded silt sized quartz mineral grains.
- (iii) There are frequently large, well rounded chert grains present.
- (iv) Most often but not invariably there is a greater abundance of soft matrix minerals - kaolinite plus hematite/limonite rather than the almost invariable $\angle$ quartz or chert cement of the Pandurra Quartzite.

Separate characteristics are noted below.

2.3.2 Main Open Pit

- (i) Mainly rounded grains plus finer quartz matrix with kaolinite but some rounded to subangular grains in equigranular bands.
- (ii) Few quartz rock fragments are seen.
- (iii) In the matrix component, kaolinite is more abundant than chert.
- (iv) Moderate amount of matrix.
- (v) Lack of heavy minerals.

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2.3.3 West Lagoon

- (i) Marked size difference between the large rounded aeolian grains component and the finer matrix quartz.
- (ii) An abundance of rounded chert grains in some specimens.
- (iii) Few equigranular quartz bands are seen.
- (iv) An abundance (approximately 5 volume percent) of clay or chert replacing clay - occurs as matrix - rather more than is seen in Main Open Pit specimens.

2.3.4 Mystery Area

- (i) Two specimens show large rounded aeolian grains with fine matrix as noted in other areas, the remaining specimen is finer grained and moderately equigranular.
- (ii) Quartz rock fragments are becoming abundant (to 10 volume percent).
- (iii) Fairly fresh microcline grains, only slightly sericitised.
- (iv) Few heavy minerals are present.
- (v) Carbonate clasts or granules are common in the matrix (Whyalla Sandstone here rests on Woocalla Dolomite).

2.3.5 Plateau Area (SW of Mill)

- (i) Rounded quartz grains, rock fragments and chert grains of aeolian origin but close packed with minimal matrix.
- (ii) The matrix is almost completely silicified so that only trains of hematite, fluid inclusions and opaque particles indicate former outlines of aeolian grains.
- (iii) Optical continuity between original quartz and $\propto$ quartz 'matrix' is maintained.
- (iv) Any original silty quartz matrix grains have been dissolved and absorbed by growing quartz aeolian grains.
- (v) Kaolinite present is showing intergrowths of muscovite.

2.4 Tregolana Shale

2.4.1 Mystery Area

A fine grained variable quartz, sericite, hematite, carbonate, kaolinite shale. The vertically changing mineral distributions delineate banding.

2.5 Corrubura Sandstone

2.5.1 Mount Gunson

A great variation in median grain size between specimens of somewhat different appearance but most quartz grains are angular to ovoid and tabular and grains within one hand-specimen are more or less equigranular. Appreciable limonite/kaolinite matrix is abundant. Much strained quartz including Böhm Lamellae is seen. Very few quartz rock fragments

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are present. Fresh microcline occurs and more than one percent of heavy minerals including zircon and tourmaline, also some flakes of detrital muscovite. Silicification was limited and patchy.

2.6 Simmens Quartzite

2.6.1 Mount Gunson

Only two specimens were collected, both are much silicified and display prominent current bedding, showing shallow water conditions of sedimentation and cross-current activity from discharging streams. An absence of soft matrix minerals (clays, etc) is noted. Conversion of edges of some quartz grains to chert show post-depositional unstable pH conditions after sedimentation. This is distinct from Pandurra Quartzite where the common occurrence of current bedding and presence of an appreciably heavy mineral suite is not seen. The quartz grains are relatively unstrained and this may indicate a different provenance from the Pandurra Quartzite.

3. DISCUSSION SUMMARY

3.1 Gunson Area

The rock sequences examined by R.K. Johns (Ref. 6) consists of Upper Proterozoic (Adelaide System) Pandurra Quartzite overlain unconformably by the Tent Hill Formation of (from bottom upwards) Woocalla Dolomite, Whyalla Sandstone and Tregolana Shale. Above the Tregolana Shale occur the Corrubura Sandstone capped by Simmens Quartzite occurring as the outlier of Mount Gunson.

The Pandurra Quartzite appears to have consisted originally of somewhat rounded sand grains as seen by hematite trains within diagenetically enlarged more angular grains, masses of which have been cemented together by quartz or chert to form large frequently angular clasts. Poor sorting, a characteristic of Pandurra Quartzite is significant for a seasonal high energy environment. Chert grains are present in appreciable numbers but not as much so as in the overlying Whyalla Sandstone. Kaolinite does form the matrix in places, where it has not been replaced by chert. The replacement of kaolinite by chert is of common occurrence in sub-greywacke sandstones according to Dapples (Ref. 1), the mechanism of the process not being clear but it is thought that clay sized interstitial material containing small particles of quartz, act as nuclei for precipitation of additional silica. Larger clasts of quartz rock material and more quartz rather than chert cement at West Lagoon compared with the Main Open Pit indicate a higher energy environment during sedimentation and a subsequent reducing environment rather than oxidizing as is seen at the Main Open Pit.

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In the Mystery/Pernatty Lagoon area a zone of red-mud containing roughly 60 volume percent of scattered large subangular quartz grains indicates an estuarine locale to a fast moving stream. Remaining sections show a very poorly sorted sand grain component with a fair concentration of quartz gneiss grains from a metamorphic rock source. An abundant matrix of kaolinite (or chert after kaolinite) and a high quartz rock to mineral grain ratio indicates an estuarine/shore line deposition of arrested stream velocity of a fast moving stream hypothesis (vide supra).

In Woocalla Quarry a greater variation in individual quartz grain size, lack of matrix and of rock fragments indicates a beach environment above normal wave action to account for the poor sorting.

Lack of fine matrix is significant with the lack of sulphides in Woocalla Quarry/Pandurra Quartzite suggesting that the sulphides are mainly present where matrix minerals are abundant as in river estuaries or channel deposits, precipitated as gels (sulphides or hydroxides) in the more alkaline saline waters of an estuary or perhaps washed down as sulphides with clay minerals from playa lakes finally settling in a zone of reduced water velocity. Lack of heavy minerals and occurrence of hematite trains within quartz grains show that many of the quartz grains have been recycled from a previous littoral setting.

The overlying Woocalla Dolomite is overstepped by the Whyalla Sandstone west of a narrow strip of hinterland on the west shore of Pernatty Lagoon, is absent in the Main Open Pit area but is seen to occur in the drill cores of the Mystery-Pernatty Lagoon '3 - 6' programme. Here the dolomite is remarkably pure and of a manganimiferous type as determined by Dr. M.T. Barnet. According to Dana

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(Ref. 8) and Deer, Howie and Zussman (Ref. 9), Cu as such is not accommodated in the dolomite lattice although Mn^{2+} , Co^{2+} , Pb and Zn will substitute for Mg (Ref. 8) and Pb will substitute for Ca in the lattice. No mention of substitution of Mg or Ca by Cu is made by either sources, so that Cu present in even small quantities will be there as a discrete non-dolomite probably sulphide phase. Some fine scattered specks of chalcopyrite are seen in some shaly beds towards the bottom of the Woocalla Dolomite at Sweet Nell but these are of very limited occurrence and are quite different from the pure dolomite which constitutes the bulk of the Woocalla Dolomite. Weathering of the dolomite, as at Woocalla Quarry gives a shaly looking black rock but apart from 2 volume percent detrital quartz, impurities consist almost entirely only of exsolved manganese from the dolomite crystal lattice as manganese dioxide. Rankama and Sahama (Ref. 10) report an average of 12.6 g/ton of Cu in dolomite which is approximately 13 ppm. Johns, King and Dickinson (Refs. 11, 12 and 13 respectively), suggest dolomite at Mount Gunson to be the source of copper but in view of the likely overall small concentration of copper in the main dolomite, lateral secretion of the latter by meteoric waters with secondary deposition in fissures in the underlying Pandurra Quartzite seems of little importance.

Whyalla Sandstone is discussed in Part I of this report where the shapes and surfaces of large spherical frosted grains were studied in deslimed percussion drill samples. These samples cover a more widespread area than the diamond drill holes and surface exposed localities. However differences of size and shape of grains and mineral content have been observed for specimens from the Main Open Pit, West Lagoon, Mystery and 'Plateau' areas. The most distinctive characteristic generally is the two distinct modes of grain size which comprise the bulk of Whyalla Sandstone exposed or drilled. Large aeolian grains

occur with frosted surfaces to several millimetres in diameter and sharp angular grains of a few microns diameter with kaolinite occur in the matrix, which is generally abundant. It is assumed therefore that the palaeo-landscape was an arid desert one, consisting of sand dunes and salt lakes. Periods of flash flooding overflowed the lakes and the temporary streams so formed incorporated dune sand particles with lacustrine mud which together drained in to channels occurring below the water table as a semisolid mass so that graded bedding did not form. Pyrite which developed from the bacterial reduction of gypsum and anhydrite in the original playa lakes was washed into the channels with the mud and dune quartz. This provided nuclei for replacement by copper solutions draining from streams which during infrequent heavy rains covered a very large part of the catchment area of this low-relief landscape. Fluctuations of water-table due to eustatic rise in the water level or isostatic rise of the shelf or craton caused solution and mobility of the sulphides and percolation through the sandstone downwards to produce supergene zones particularly in the underlying Pandurra Quartzite where mobility was restricted by secondary quartz or chert, sealing-off the porous structure of the erstwhile sandstone. Lowering of Eh resulted in further bacterial reduction of sulphates to produce further pyrite nuclei for copper sulphide precipitation.

Differences between the various sampled Mount Gunson area, Whyalla Sandstone occurrences are slight.

The presence in the Main Open Pit of bands of superpose more equigranular quartz of subangular aspect indicate a eustatic rise in the water level above the drainage channels containing washed-out desert dune sand to wash in beach dune or beach sand (MacCarthy - Ref. 14). The presence of moderate matrix suggests less mixing with playa lake material, the waters draining to the flood channels only skirting playa

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lakes. The gross oxidation of exposed deposits suggests a higher water table existed when the copper deposits formed than occurs today or possibly uplift of the area may have occurred.

The West Lagoon orebody occurs on the shores of Pernatty Lagoon and it being roughly level with the present water-table is undergoing supergene sulphide enrichment from percolating cupriferous waters. The lower situation of West Lagoon as compared with the Main Open Pit equates with a greater clay matrix component washed-in from previously existing playa lakes (cf. Pandurra Quartzite, Mystery Area). A somewhat different provenance is shown by the abundance of near spherical chert grains (of desert dune origin ?).

Mystery area Whyalla Sandstone contains large and small clasts of carbonate originating from 'underlying' eroded Woocalla Dolomite and also an abundance of rounded quartz rock fragments indicating that sand dunes of varying quartz lithologies provided material for these deposits and that periodical flooding was extensive and covered a wide area. The presence of microcline feldspar further confirmed the normally arid climate which prevailed over the catchment area.

The Plateau area south-west of the Mill Site is of rather high elevation comparatively speaking and here silicification of the sandstone has been most complete because of the paucity of original 'kaolinite' matrix. The sandstone consists for the main of tightly packed spherical sand grains of frosted surface, of aeolian origin and with occasional kaolinite pseudomorphs of original microcline. Some of the kaolinite is developing into muscovite which indicates an increase in pH (5.5 to 7.5) and slight increase to a positive Eh, Dapples (Ref. 1), a change which has taken place since kaolinisation of

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feldspars and probably due to a fall in the water-table to the present day. Incorporation of playa lake mud was negligible and concurrent absence of sulphides is significant.

3.2 A Comparison with Other Red Bed Copper Mineralised Areas

A present day example of sulphides accumulating under continental conditions on the east African coast is cited by Garlick (Ref. 15), where tropical lagoons behind a protecting coral barrier reef are bottomed by appreciable organic material which at low tide give off a smell of H_2S and the muds are seen to contain mud-fine iron sulphide. Stagnant pools are dammed by sand dunes. Johns (Ref. 6) makes an allusion to the Gunson sedimentation environment which is similar to that cited in east Africa except for the existence of Collenia reefs rather than coral reefs.

Garlick (Ref. 15) showed how the Nile delta received 51×10^6 metric tons of suspended solids and 21×10^6 tons of salts annually. Assuming 100 ppm of Cu as an average figure for the sediments and igneous rocks in the Nile drainage area, 7,200 tons of Cu would be carried annually to the sea. So that after 1,000 years of erosion a 7×10^6 ton copper deposit might result. However Garlick pointed out that to ensure the concentration of an orebody it was necessary for there to exist,

- (a) a means of efficient and selective precipitation of copper from weak solutions and
- (b) a method of keeping much of the 51×10^6 tons of detritus away from the precipitation area.

Garlick goes on to say that the precipitation process can occur in stagnant sea bottoms where H_2S is generated in large volumes and will precipitate most or all of the copper before iron is precipitated in major quantity.

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Garlick (Ref. 16) states that the literature on stratiform disseminated sulphide deposits in arenaceous sediments is somewhat meagre. He discusses the arenaceous mineralised horizons at Mufulira and states that arenites are not typically deposited under anaerobic conditions since cross-bedded arenites demand fast flowing water which is usually oxygenated, deposited rapidly and remains permeable for long periods after burial. This observation indicates that the mineralisation at Gunson would therefore have had to be flushed in as sulphides or sulphide gels from a lake bed where anoxic conditions prevailed and/or deposited on suitable nuclei within the sandstone matrix after sedimentation in drainage channels adjoining the water-table. However in other ways there are broad similarities between the Mount Gunson and Mufulira mineralisation environments, with the presence of :-

- (a) feldspar or pseudomorphs after feldspar in the sandstone/quartzite,
- (b) a supposed mainly arid climate,
- (c) sulphidite layers,
- (d) load casts,
- (e) anhydrite concretions,
- (f) barren aeolian arenites and mineralised aqueous arenites, and
- (g) algal growths in neighbouring dolomite.

Differences between the environments are shown by the following characteristics at Gunson :-

- (a) absence of mineralised argillite horizons,
- (b) absence of cross-bedding ,
- (c) silicification of much of the Pandurra Quartzite,
- (d) narrowness of mineralised horizons and their horizontal discontinuity, their being centred at largely past and present water-table levels,
- (e) present day slow formation of sulphides in salt lakes,
- (f) absence of metamorphism and tectonic structures to mobilise ore.

Garlick prefers to support diagenetic mineralisation (as distinct from syngeneses within neighbouring argillites) in the arenite sequence because of the porous nature of these beds (at Mufulira). He states that it is possible that contemporaneous diffusion of mineralising solutions through permeable sand beds may have taken place en route to the basin where the major proportion of metals was precipitated syngenetically. Much of the mineralisation at Gunson would bear a diagenetic interpretation. Garlick again states that the source of sulphur was biogenic (from sulphates) and source of metals the erosion of an extensive land surface as has been suggested for Gunson.

Brandt (Ref. 17) suggests that a parallel may be drawn between the Mufulira deposit as interpreted by himself and also White Pine, Michigan as interpreted by White and Wright (Ref. 19) where the copper is syngenetic derived from waters in which the associated sediments were laid down and precipitated in stagnant lagoons enclosed by levies and sand bars in a delta topography. Copper bearing siltstones were deposited in lagoons between separate distributory streams and the mineralisation being terminated by inundation of the bars and levies and filling-up of the lagoons with barren sediment. The parallel with Mufulira is very close and resemblances to Gunson are noted. Brandt goes on to quote Stanton (Ref. 20) who, discussing numerous North American sulphide deposits in the Appalachian belt in sedimentary-volcanic terrains, described them as biochemical precipitates formed in shallow-water offshore lagoons in seaboard volcanic areas, fumarolic activity being necessary to supply metals and sulphur to sea water, Stanton did not consider that metals in solution could have been supplied in sufficient concentrations by erosion processes only. However a ready-made volcanic source for copper and sulphur does not exist in the Zambian Copperbelt, White Pine or in the Gunson area. In the absence of any evidence for volcanic or

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fumarolic activity it must be assumed that the metals, if syngenetic, were derived originally from basement rocks by erosion. Even though geological opinion is generally unfavourable to the idea of erosion being competent to supply large amounts of metal the probability of this occurring is examined by Garlick (Ref. 16) as cited above, and discussed by Gilmour (Ref. 21) who describes several present day examples of copper yielded by the weathering of pyritic orebodies, being carried in solution by streams and precipitated in bogs fed by streams. Present day organic sequestration of copper being concentrated in peat bogs is discussed by Fraser (Ref. 22), where concentration of up to 10 percent copper is reported in south-eastern New Brunswick, Canada. The copper is leached from country rock (boulder clay ?) and emerges as seepages in clearings in the bog.

A similar geology and palaeogeography to that of Mount Gunson has been described for the Lower Roan Formation at Chibuluma on the Zambian Copperbelt by Whyte and Green (Ref. 18). The mineralisation here is in aqueous micaceous arenite. The presence of a great deal of pyrite and carrollite at Chibuluma suggests precipitation of part of the sulphides in a neutral or alkaline environment whereas copper, bismuth and silver at Gunson were precipitated from acid waters and the climate was possibly seasonally less arid. Palaeogeography indicates a river channel most likely cutting into thick aeolian sediments prior to a marine transgression and then submerged. The channel was present but submerged throughout deposition of quartzitic arenites of the ore horizon and shallowing of the water from time to time exposed a sand bank. Facies complications which occurred in the channel were due in large part to scouring and mixing of sediment by current activity in a southern extension of the submerged channel. An alternative suggestion by Green

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for the origin of the oreshoot was deposition in a playa lake, in which case the proposed marine transgression did not take place until after the close of ore deposition. It was suggested that the basin was periodically unindated with flash flood waters carrying Cu, Fe and Co leached from the surrounding country. The metals were probably deposited as sulphides under anaerobic conditions with gradual evaporation of water in the basin but incomplete drying out to account for facies changes, scour channels and stratification of copper, cobalt and iron (pyrite) facies. Lack of evidence of sun-cracks suggests the lake never completely dried out. Other interesting observations made were that the high grade mineralisation is confined to the hard compact quartzite arenite; also that syngenetic deposition in the orebody could have taken place as -

- (a) chemical precipitation as hydroxide,
- (b) chemical precipitation as sulphide, or
- (c) detrital deposition of metalliferous minerals.

The geology, palaeogeography and interpretation of the unrolled plan of geology, bearing in mind that the Chibuluma West Orebody was metamorphosed, folded and faulted is very similar to that of Mount Gunson. However as in all the previous examples of sulphides developing in arenite, there always appear to be neighbouring mineralised argillaceous rocks which provided the syngenetic aspect of sulphide development. These mineralised argillites do not exist at Gunson to supply mineralisation for the orebodies which are mainly at the Whyalla Sandstone/Pandurra Quartzite contact.

The various copper deposits of the European Upper Permian Zechstein Sea (Oberc, J. and Serkies, J. - Ref. 2) show biogenic syngenetic development in a reducing medium of sea water and silts containing organic matter (ex plankton) which produced the mineralised bituminous Kupferschiefer with associated underlying sandstones which also contain ore.

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To account for copper in porous sandstones, post-depositional processes were suggested to have taken place on a large scale. Because of the paucity of copper in sea-water, weathering of pre-existing endogenic deposits in the surrounding land areas and circulating ground waters in underlying Lower Permian sandy sediments were involved, to supplement the copper already dissolved in sea-water. A direct ratio was established between copper and organic carbon present in the Kupferschiefer and topmost White Zechstein sandstone, the carbon being derived from dead plankton; lead and zinc occurs in the upper part of the deposits. One interesting observation which bears comparison with Gunson are the occurrence of maximum concentrations of copper in a very limited vertical section of the upper White Zechstein sandstone and overlying copper shales and this is purported as evidence of a very rapid supply of large quantities of copper to the Zechstein Basin and rapid precipitation of that copper during the relatively short period of sedimentation of copper-bearing sediments directly after transgression of the sea. Zechstein deposits were considerably enriched in sulphates and these were reduced to supply sulphur for sulphide production. No mention of sulphur reducing bacteria is made. Adsorption of Cu ions on sinking particulate matter in the sea was thought to be the mechanism for bringing the copper to the reducing zone. Sulphide bodies were formed by diagenetic differentiation of the more soluble sulphides from lithifying bottom sediments. Decrease of pH during diagenesis favoured changes in trace element mineralogy so that bismuthinite and native silver were precipitated. Possibly similar conditions existed for the occurrence of silver and bismuth sulpho-salts at Gunson. The White Zechstein sandstone as mentioned above is in places impregnated with copper, mainly as chalcocite with some chalcopyrite (as for Gunson - Pandurra Quartzite/Whyalla

Sandstone contact). The essentially oxidizing sedimentation medium of the White Zechstein Sandstone as compared with the nature of mineralisation found in it is evidence that the metallic minerals were emplaced by later downward percolation of sulphide solutions in part resulting from diagenetic differentiation in the overlying copper bearing shales (Kupferschiefer) and the copper in the sandstone is therefore epigenetic. Decolorisation of this sandstone extends to greater depth than does the sulphide impregnation (see West Lagoon, A4004, Whyalla Sandstone, Ref. 2). Supergene enrichment in joints and fissures are regarded as post-diagenetic by weakly oxidizing secondary solutions circulating the deposit, (the similar processes at Mount Gunson are described as diagenetic since it is difficult to distinguish these processes from diagenetic changes occurring within the Whyalla Sandstone or Pandurra Quartzite.

Other worldwide areas of metalliferous mineralisation of arenites are rare and Garlick (Ref. 15) mentions only two, other than the Zambian Copperbelt occurrences (Brown - Ref. 24 and Grip - Ref. 25).

Above the Whyalla Sandstone of the Tent Hill Formation of Gunson rests the relatively thin Tregolana Shale (<50 feet in the Gunson area : Johns - Ref. 6) and unlike its equivalent, the Kupferschiefer of the German Zechstein sequence (Gignoux - Ref. 26), it is largely non-cupriferous and is not bituminous. Lack of bitumen is equitable with lack of abundant life forms in Upper Proterozoic times. Tregolana Shale consists of very variable concentrations of fine sedimented quartz, sericite, dolomite and kaolinite with variable concentrations of hematite particle inclusions giving a banded appearance to the rock.

Above the Tregolana Shale, the Corruburra Sandstone and Simmens Quartzite occur in the area as an outlier - Mount Gunson, which rises mesa-like above the gently undulating plateau/salt lake topography.

The Corruburra Sandstone bears some resemblance to the Whyalla Sandstone below, but the larger quartz grains are ovoid to rounded-tabular in shape and have all the appearance of being water worn. The grain size is very variable at different horizons but equigranular along bedding planes. The rock is feldspathic containing detrital fresh microcline and unlike the arenaceous deposits below, it contains an appreciable heavy mineral content. The rock is somewhat similar to the Feldspathic Quartzite (occasionally mineralised) of the Zambian Copperbelt and similarly is above the main ore horizon.

The overlying Simmens Quartzite is somewhat coarser grained than the Corruburra Sandstone, and almost completely silicified. The feldspar content is quite high at 4 percent in the specimen examined and heavy mineral content at 2 percent is high for 'sandstone' rock types in the Gunson area. Current bedding is very marked showing shallow water deposition.

The climate in both Corruburra Sandstone and Simmens Quartzite times remained arid. The deposits are shallow water shelf/estuarine, which lay dormant during lengthy arid spells punctuated by sporadic flash flooding of rivers. Similar to the conditions applying today, in fact. A relative abundance of heavy minerals and fresh feldspar indicates a source of rock material which has not suffered as much recycling and mineral differentiation as the underlying sandstones.

3.3 Fluid Inclusions

Fluid inclusions are abundant in quartz overgrowths and planes of weakness in mineral grains mainly, but also patchily in rock fragments and chert grains. Unfortunately only once was a halite crystal seen in an inclusion to prove the salinity of the contained solution. However, it is assumed that apart from a few primary inclusions present in primary quartz grains before diagenesis, most of the inclusions contain the brine which transported metal values resulting in the present copper deposits. It is seen that the inclusions were trapped during silicification by the silica-rich brine solutions. No fluid inclusions could be seen in the sulphides (these are difficult to see in opaque ores), although scattered fluid inclusions are seen in the malachite of the Pandurra Quartzite of the Main Open Pit. Penetration along fissures and grain boundaries by the brine is seen in stressed grains and in adjoining mineral grain boundaries in rock fragments. Fluid inclusions in chert fragments were presumably formed during crystallisation of the chert in some previous epoch.

The dearth of fluid inclusions in the Woocalla Dolomite indicates its formation by primary deposition and slow compaction during sedimentation with little or no circulation of brine after compaction.

It was stated by Roedder (Ref. 27) that fluid inclusions in most cases do not lose or gain appreciable quantities of material after being trapped. Therefore for the purpose of interpretation of ore genesis some idea of the composition of the original ore forming fluid could be obtained by using methods such as Roedder E, Ingram B, et al (Ref. 28). Determination of filling temperature should also be done, to prove the low temperature of the brine at the time it was trapped in the void.

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4. DESCRIPTIONS OF SECTIONS

4.1 Pandurra Quartzite/Main Open Cut

A 4252 (D.H. 3-4-1D, 21.9m)

Bands of alternating fine and coarse quartz grains. The bands vary between rounded and subangular, poorly sorted grains and well sorted bands of larger rounded grains. The quartz mineral grains vary between 0.15 and 2.1 mm with the average varying between .25 and 1 mm within different bands. Most quartz rock fragments are close to 2.5 mm diameter and a few of these are gneissose in appearance. Unusually, rock fragments are composed of equigranular mosaics of quartz mineral grains, the individual grain sizes being close to 0.05 mm. A minor proportion of quartz grains are fused together and a narrow quartz selvage occurs at the mutual boundary. Trains of hematite particles and fluid inclusions (approximately 1μ) occur just outside primary quartz grain boundaries indicate a more rounded original grain. Abundant kaolinite (locally modified to sericite) plus hematite which was present interstitially has now been replaced by quartz or by chert. As a proportion of matrix kaolinite is equally abundant in fine and coarse quartz bands but chert is less frequently seen in the finer bands. In the coarse bands the kaolinite is seen mainly in vugs of probably decomposed feldspar of variable size and up to 1 mm dimensions. Grains of chert to 0.5 mm are seen in association with the kaolinite and may be an example of chert replacing clay (Ref. 1).

Mineral modal volume percent :-

quartz mineral grains	87
quartz rock fragments	3
chert	2
kaolinite (locally modified to sericite)	7
hematite	< 1
sulphides (chalcocite and chalcopyrite)	1

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A 4254 (D.H. 3-4-2D, 18.0m)

Similar to A 4252 in grain sizes, distribution of minerals and poor sorting of mainly finer sizes. One band of larger grains, including quartz mineral, quartz rock and chert show a number of fairly well rounded grains but mainly angular to subangular grains and much kaolinite (altering to sericite) matrix. Quartz mineral grains vary between .05 and 1.2 mm; and rock fragments vary up to 3.5mm; chert grains vary between 0.7 and 1.3 mm. Complex suturing occurs at boundaries between quartz grains in some quartz rock fragments. Diagenetic alteration of matrix (clay to quartz) is patchy. Scattered hematite flakes occur in clay and chert in the matrix. Occasional large (.10 to .20 mm) worn detrital zircon and some messy sphene are seen.

Modal volume percent of mineral composition :-

quartz mineral	87
quartz rock	5
chert fragments	1
chert matrix	2
kaolinite (altering to sericite)	5
zircon and sphene	<1
hematite	<1

A 4256 (D.H. 3-4-3D, 8.5m)

Poorly but still better sorted than A 4252, 4254, with fine and coarse bands and diagenesis has resulted in the replacement of much of the clay matrix by quartz addition to quartz (mineral grains and rock fragments). A little orthorhombic chalcocite (shows 001 cleavage), altering along cleavages to atacamite. Elsewhere fine fractures (< .1 mm) traverse quartz grain boundaries, parallel to bedding and are filled with malachite and atacamite. This is an oxidation effect and probably has something to do with the greater volume occupied

by the oxide mineral which has replaced the pre-existing copper sulphide.

Occasional round zircons (approximately .05 mm) are seen in the interstices and some small amber coloured rutile prisms.

Judging by the mutual replacement of atacamite by secondary quartz and vice versa it would appear that diagenesis has continued alongside oxidation of the sulphide minerals. Some fractures across quartz grains are part filled with atacamite intergrown with secondary quartz.

Strain is shown by only a minority of quartz grains and these apparently do not occur in zones adjoining mineralisation (?). Diagenesis by silica solution has been extensive but chert has not precipitated. Very little clay mineral remains in the interstices. Hematite trains and fluid inclusions adjoining primary quartz grains indicate the extent of quartz grain growth, the quartz grains being originally more rounded as a rule. Joins between quartz grains show simple suturing or somewhat sinuous bands of secondary quartz a few microns only in width.

Modal volume percent mineral composition :-

quartz mineral grains	98
quartz rock fragments	2
chert	< 1
kaolinite	< 1
zircon	< 1

Quartz mineral grains average 0.5 mm; vary from 0.05 to 2.1 mm. Quartz rock fragments are variable in size to 3 mm.

A 4257 (D.H. 3-4-3D, 14.0m)

Similar to previous sections in this series, coarse more equigranular bands showing some rounded grains and finer bands with a proportion of much larger subangular 'erratics'.

A minimum of matrix is seen between the quartz grains which show sinuous bands of secondary quartz between grains, a few microns wide. Original grain boundaries are demarcated by hematite particle and fluid inclusion traces. Optical continuity is exhibited by the secondary growths away from the contacts. Some residual kaolinite remains in pockets (ex-feldspar ?) but some remains unreplaced in channel ways between the grains. Little or no chert replacement of clay is seen. Some large round chert grains occur and quartz gneiss fragments are uncommon. Tabular quartz mineral grains with rounded edges (water-worn) average 0.24 mm, vary from 0.05 to 0.80 mm; quartz rock fragment are usually large and vary up to 0.80 mm.

The kaolinite is occasionally brownish and shows a higher than normal birefringence; it is possibly altering to brownish sericite.

Modal mineral composition of the rock :-

quartz mineral grains	96
quartz rock fragments	2
chert	< 1
clay	2
hematite	< 1
zircon	< 1

All four Pandurra Quartzite specimens are very similar, the variations amounting to differential diagenesis by silica containing solutions. Replacement of kaolinite by chert is the predominant change rather than secondary

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quartz growth on to primary grains. This is discussed by Dapples (Ref. 1) who suggests that 'clay mineral aggregates may be completely replaced by chert. More stable quartz may be precipitated on primary quartz grains as overgrowths and fillings of interstitial space precipitated when the silica content of the water in the sandstone aquifer is no more than several parts per million and the pH is ± 7 with a general prevalence of oxidising conditions. For these conditions to prevail the possibility of sulphides forming con-currently with the process of silica diagenesis is not possible and therefore the latter post-dated the sulphide mineralisation.

4.2 Main Open Pit : Specimens Collected - July 1971 (Ref. 2)

A 4012, 4013 (Specimens removed from walls of the excavated Open Pit)

The quartzite samples are similar to those taken later from drill holes except that in the 'pit' samples the finer bands which are equigranular contain no large quartz clasts. However as in the drill hole specimens the large grains in coarse bands vary from rounded to subangular in shape. Matrix of the rock is chert in most unmineralised rock, though examples of quartz growth followed by complex suturing (indicating pressure solution) between the grains is seen. In the mineralised specimens chrysocolla is the main matrix mineral and it is apparently being replaced by malachite and atacamite which are also replacing quartz grains in part or in whole. Chert/chalcedony growths have worked their way along secondary quartz margins where quartz grains were previously cemented by secondary α -quartz.

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4.3 West Lagoon : Specimens Collected - July 1971 (Ref. 2)

A 4002, 4005, 4009 and 4010

Similar to Main Open Pit drill hole specimens, with alternating fine equigranular bands containing scattered large clasts of quartz rock and coarse bands of quartz subangular to somewhat rounded grains with abundant matrix of clay mineral. However much of the quartz consists of quartz mosaics with slight suturing or a sinuous secondary quartz 'join' between individual quartz grains. No chert is seen.

The precipitation of chert was therefore related to alkaline, oxidation conditions and bearing in mind that these West Lagoon rocks contain only sulphides, secondary quartz (as distinct from chert) deposition was due to deposition from slightly acid solution and slightly negative Eh.

4.4 Mystery/Pernatty Lagoon Area

A 4238 (D.H. 3-6-1D, 29.4m)

In hand specimens the rock consists of large round and angular grains of quartz with abundant red-brown tough matrix of somewhat greywacke appearance.

Quite different from other specimens of Pandurra Quartzite examined, the matrix is seen to consist of fine angular quartz fragments averaging around 0.05 mm, varying between 0.016 and 0.11 mm; hematite flakes (average 0.030 mm); and fine chert. Hematite flakes form narrow zones around larger quartz rock fragments or as laths (approximately .08 mm) otherwise intermixed with the cherty substance.

The large grains of quartz are angular, show much irregular cracking and are strained; they average 0.32 mm

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but vary up to 0.80 mm. Large chert grains, rounded to angular and of similar grain size to large quartz rock fragments occur, up to several mm's in size.

A structure originally thought to be foraminiferal, (four joined rounded chambers, 0.5 mm diameter) with a thin cherty test and filled with fine chert, is present. Dr. D. Burger, Micropalaontologist, of the Bureau of Mineral Resources has looked at this and though he cannot explain the existence of the structure he thinks it may not be of organic origin.

Modal composition, volume percent :-

quartz mineral	10
quartz rock	48
chert - matrix	30
chert - grains	3
kaolinite matrix	2
kaolinised feldspars	2
hematite	3
opaque minerals	2

A 4241 (D.H. 3-6-2D, 43.6m)

A poorly sorted quartzite, grain sizes not very different but perhaps somewhat coarser than previous specimens observed from the Main Open Pit. Banding by finer grained quartz is not evident.

The quartz grains are mainly subangular but mineral grains and rock fragment of rounded aspect are seen and occasional angular quartz rock fragments. The quartz mineral grains average 0.64 mm, vary between 0.08 to 1.1 mms; quartz rock fragments generally vary between 0.80 mm to 2.3 mms. A small number of angular raft-like grains of sheared quartz gneiss are seen.

The matrix of kaolinite altering to sericite (or illite ?) is abundant. Alteration to sericite appears to take place mainly where the matrix adjoins large quartz grains but sometimes also within the clay matrix, sericite forms concertina book forms of some size. Hematite flakes patchily coat quartz fragments. Worn detrital zircon grains occur in the matrix from 0.016 to 0.080 mm's.

Modal mineral composition, volume percent :-

quartz mineral grains	27
quartz rock fragments	60
kaolinite	10
sericite	2
hematite	1
zircon	<1

A 4245 (D.H. 3-6-3D, 58.1m)

A poorly sorted quartzite and similar to A 4241 except for the higher proportion of quartz rock fragments in the latter and for replacement of kaolinite matrix by chert. The quartz grains are mainly subangular but with a high proportion (approximately 20 percent) of angular grains and a few rounded quartz grains of intermediate size. The size ranges are - quartz mineral grains 0.2 to 3 mm, average 0.5 mm; quartz rock fragments 0.5 to 1.5 mm. Much of the original kaolinite has been replaced by interstitial chert. Large spaces between the quartz grains are filled with chert and clay; quartz grains have patchily become welded together by narrow sinuous joins of secondary quartz. Many quartz mineral grains appear to be finely and irregularly cracked, but strain is not well marked and many grains show no strain characteristics. Heavy minerals are absent. Scattered vermicules (vermiculite or other clay mineral - to 0.48 mm long 'book' forms occur in the matrix, some have interleaved hematite - possibly altered biotite.

A proportion of chert grains contain opaque matter, mainly angular grains (approximately 0.5 mm, to 0.8 mm).

Modal volume percent mineral composition :-

quartz mineral grains	78
quartz rock fragments	3
chert grains	2
chert matrix	8
kaolinite	4
hematite	5
vermiculite (?)	< 1

A 4250 (D.H. 3-6-4D, 48.6m)

Similar to A 4241 and 4245; poorly sorted mainly angular to subangular quartz mineral grains and rock fragments with occasional large rounded grains, unbanded and abundant matrix. However much of the matrix in this rock as in A 4245 is cherty it having replaced original kaolinite. Quartz mineral grains and rock fragments vary over a similar size range, between 0.1 and 1.5 mm, average 0.6 mm, except for scattered large quartz rock clasts, around 3 mm. Fine chert has partially replaced coarse quartz grains around edges but many of these grains are welded together by a thin sinuous layer of secondary quartz. Chertification would appear to post date secondary quartz cementation. Fine particle hematite, speckles primary chert grains and secondary chert matrix.

Subangular chert grains (approximately 0.7 mm) are scattered around the section.

Occasional interstitial subhedral chalcopyrite to 0.24 mm maximum size. No other heavy minerals are present.

Modal mineral composition as volume percent :-

quartz mineral grains	64
quartz rock fragments	10
chert grains	3
chert matrix	17
kaolinite/sericite	3
hematite	2
chalcopyrite	1

4.5 Woocalla Quarry

Specimens collected from the quarry have variable angular, subangular and rounded quartz grains, some of this variability being due to secondary silicification during diagenesis. Some specimens show fine grain bands; others are cherty - fine and coarse grained types. They have been subdivided as follows :-

4.5.1 Fine grained, cherty, subangular to angular grains, banded (according to grain size). Some large quartz clasts. A 4225.

4.5.2 Fine grained, cherty, subangular to rounded grains. A 4227.

4.5.3 Fine grained, cherty, subangular to angular grains. A 4228.

4.5.4 Coarse to fine grained, very cherty, unsized subangular to rounded grains. A 4226.

4.5.5 Coarse to fine grained, slightly cherty, unsized subangular to rounded grains. A 4229, 4230 4232.

A 4225

More or less equigranular, subangular to angular grains for different bedding horizons with sharp cut-off between bands, except for occasional large subangular quartz erratics. Growth of the quartz grains is indicated by trains of hematite particles and fluid inclusions ($1-2\mu$) adjoining the primary grain boundaries which generally show the pre-diagenetic grains to have been more rounded. Quartz growth is in optical continuity with the primary nuclei. About two percent of quartz grains exhibit strain. Extensive patches of quartz mosaic occur where secondary silicification has fused quartz mineral and rock fragments together with sutured contacts. Average grain size for different bands is 0.16, 0.32 and 0.64 mms; range 0.1 to 2 mm. Erratic clasts vary to 2 mms. Rock fragments comprise much of the larger grain sizes. Pockets of fine chert which occur patchily have replaced original kaolinite but leaving intermixed hematite flakes intact. Occasional muscovite flakes, up to 0.3 mm are seen in the matrix where chert has not yet replaced the softer matter. A little detrital tourmaline and zircon is seen embedded in diagenetic quartz.

The Liesegang Rings as seen in hand specimen are indicated in thin-section by rhythmic precipitation of fine hematite flakes in the matrix uncontrolled by 'bedding' planes and occur where exists a greater hematite: chert ratio.

Modal volume percent mineral composition :-

quartz mineral grains	92
quartz rock fragments	2
chert matrix	3
hematite matrix	3
tourmaline, zircon, muscovite, opaques	< 1

A 4227

Subangular to rounded quartz mineral grains and some quartz gneiss rock fragments, range 0.05 to 0.72 mm, average 0.40 mm. A proportion (approximately 3 volume percent of grains) are flattened with rounded ends. Occasional grains contain fine rutile needles in a rough trigonal pattern. Most of the quartz was originally welded by secondary quartz in a mosaic with straight or only slightly curved grain boundaries. However in remaining interstitial pockets, chert developed (having replaced residual kaolinite ?); and chert has since partly replaced grains of quartz growing in remaining channels in the quartzite. The commonly seen narrow trains of hematite particles and fluid inclusions adjoin primary quartz grain boundaries indicating the extent of secondary addition of quartz. Pressure-solution boundaries between the quartz grains are seen in A 4226 for example, are not so obvious here. However more than 50 percent of the quartz grains show stress and the grains displaying this are usually mosaics of grains in the section which impinge one on another.

Pockets of chert after kaolinite, occur between the quartz grains from 0.24 to 1 mm diameter. Occasional pockets of kaolinite as 'books' of minute flakes remain.

Quartz rock fragments tend to be more angular than quartz mineral grains of similar size.

Occasional flakes of muscovite are seen; one is seen to be embedded in secondary quartz (.08 x .03 mm).

One small area of anhedral carbonate (approximately 0.06 mm square) is seen in the interstices associated with hematite flakes.

No heavy minerals are seen.

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Modal volume percent of mineral constituents :-

quartz mineral grains	91
quartz rock fragments	2
chert grains	2
chert, interstitial	3
hematite	2
kaolinite	< 1
carbonate	< 1
muscovite	< 1

A 4228

Fine grained, somewhat cherty subangular to angular mineral grains of quartz and rock fragments (some gneiss), varying between 0.05 and 1.5 mm, average 0.24 mm. Welding of the mineral grains together by precipitation of quartz from percolating silica solutions has taken place. Some interstitial kaolinite occurring in channels up to 0.24 mm in width has not as yet been replaced by chert but primary chert fragments are in evidence. Hematite flakes are in abundance with kaolinite where present. In this specimen hematite is largely adhering to the surfaces of individual quartz grains so that secondary growth on the quartz grains was limited. Where grains have fused together to form mosaics the hematite flakes are seen to be just inside the new grain boundaries. As before strain shadows in quartz grains are most noticeable where impection of grains one on another is greatest. Chert grains are normally somewhat rounded and vary between 0.1 and 0.2 mm's, they are here mainly free of hematite inclusions. Some chert is seen to be replacing clay mineral in the interstices and this frequently engulfs flakes of hematite.

No heavy minerals are seen.

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Hematite banding here follows bedding and does not indicate Liesegang Rings showing that the latter is strictly a diagenetic phenomenon.

Modal analysis mineral composition as volume percent :-

quartz mineral grains	84
quartz rock fragments	5
chert grains	1
chert matrix	<1
kaolinite	6
hematite	4

A 4226

A coarse grained rock of subangular to rounded quartz grains 0.10 to 1.6 mm, average .64 mm and similar sized angular quartz rock gneiss fragments, occurring mainly as fused mosaics of quartz with simple straight or slightly curved grain boundaries. Secondary quartz occupying original channel-ways is indicated by trains of hematite particles and fluid inclusions (1-2 μ) demarcating a normally more rounded primary grain nucleus in optical continuity with the secondary quartz exterior. Chert grains, subangular, average 0.30 mm are clear of opaque matter. Chert also occupies spaces between quartz grains and is intergrown with and encroaching on the quartz. Rare muscovite flakes are seen trapped in secondary quartz. Traces of carbonate are trapped between interstitial chert and quartz grains and form part of the matrix.

Modal volume percent mineral composition :-

quartz mineral grains	93
quartz rock fragments	3
chert grains	1
chert matrix	2
carbonate matrix	<<1
muscovite	1

A well silicified Pandurra Quartzite.

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A 4229

Individual grains of quartz which were more rounded before diagenesis as indicated by included hematite grains, range 0.04 to 1.2 mm, average 0.40 mm, many grains somewhat flattened and oriented to bedding. Sporadic strained quartz grains with a $2V \sim 15^\circ$. Angular quartz grains to 1 mm, rounded chert grains up to 1 mm. Interstices where not filled with secondary quartz contain very little interstitial chert and only traces of opaque matter. One large quartz erratic present, subangular, 5 mm x 3 mm. A little muscovite and zircon are present.

Modal volume percent mineral composition :-

quartz mineral grains	91
quartz rock fragments	7
chert grains	1
chert matrix	<1
muscovite, hematite, zircon	<1

A 4230

Similar to A 4299 except for generally larger quartz rock fragments (to 1.6 mm). Quartz mineral grains range from 0.10 to 1.03 mm, average 0.5 mm and many are flattened along bedding planes. Only the matrix is appreciably cherty and traces of muscovite occur.

Modal volume percent mineral composition :-

quartz mineral grains	94
quartz rock fragments	2
chert grains	1
chert matrix	3
muscovite	<1
hematite	<1

Hematite may be secondary after sulphides but is 'primary' in the matrix and in chert.

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A 4232

Similar to A 4229 and 4230 except for larger grain size, more rounding of grains, a greater proportion of quartz greiss rock fragments and the presence of some heavy minerals. Quartz varies from 0.08 to 2.5 mm (including rock fragments), average size 0.5 mm. Completely silicified by secondary quartz growths and some chert segregated in remaining areas between corners of grains. Large grains show strain shadows. Chert grains vary to 0.80 mm, are angular. Scattered detrital zircon and tourmaline are now embedded in secondary quartz.

Modal volume percent mineral composition :-

quartz mineral grains	79
quartz rock fragments	15
chert grains	2
chert matrix	3
zircon, tourmaline, sphene	<1

4.6 Woocalla Dolomite

A 4233 (Woocalla Basal Dolomitic Shale, Sweet Nell)

The hand specimen indicates a fair amount of definitely 'specky' chalcopryrite and possible bornite and chalcocite present in a freshly exposed fracture across the bedding. Polished section shows opaque sulphides to be specks of chalcopryrite of size range approximately 0.20 mm down to 0.02 mm and are squarish to irregular in shape, some contain inclusions of pyrite. The sulphides have recrystallised as secondary mineralisation since they have mutual boundaries with coarse crystallised secondary quartz in laminae.

In this section the rock is seen to contain fine grain dolomite crystals alternating with opaque matter, much of which is hematitic/manganiferous, although

some subhedral sulphides are also seen. The grain size of the dolomite averages about 0.01 mm, and of the opaque subhedral minerals around 0.02 mm. Stringers of quartz as mosaics (individual grains vary 0.16 to 0.5 mm) associated with similar size clear carbonate grains which are probably secondary. Some grains of detrital quartz are also present.

Modal volume percent mineral composition :-

dolomite (fine grained)	74
dolomite, recrystallised (coarse grained)	2
quartz	4
wad/hematite	20
chalcopyrite	<1

The above specimen is a sulphide bearing horizon which is possibly the original source of part of the copper in the Pandurra Quartzite/Whyalla Sandstone where it has been precipitated after leaching from the dolomite. Before weathering the rock consisted of a fine grained manganiferous dolomite (Ref. 4).

4.7 Woocalla Quarry

A 4231

A fine granular dolomite, brown stained (average 0.01 mm in size) contains many scattered patches of manganese wad, some exhibiting fine opaque needle-like growths and patches of limonite. Clear recrystallised and coarser (average 0.3 mm) rhombs of carbonate have narrow zones of exsolved limonite particles around edges of individual grains or along cleavages. Particle limonite is scattered throughout the fine primary dolomite.

Fragments of detrital quartz (to 0.08 mm) also occur scattered through the rock.

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Disposition of manganese wad dendrites, some carbonate euhedra and cherty quartz indicate the presence of crude bedding planes. Thin sub-parallel fissures (0.05 mm width) in the rock contain clear secondary carbonate and 'recrystallised' wad, alternately in the same fissure. Most fissures parallel the bedding planes but a minor number cross-cut the bedding.

It is suggested that the dolomite recrystallised under the influence of percolating meteoric waters to reject Mn as MnO_2 .

Modal volume percent composition :-

primary dolomite	82
secondary dolomite/calcite	10
wad	5
limonite	1
detrital quartz	2

Fine fissures which developed along and across bedding of the dolomite demonstrate an easy method of transportation of metal values in solution through or from the dolomite. Solution of manganese bearing dolomite with recrystallisation to clear manganese-free carbonate shows that solution and recrystallisation in situ after release of base metal values occurred.

4.8 Pernatty Lagoon/Mystery Area

A 4237 (D.H. 3-6-1D, 13.4m)

A pale cream, fine-grained dolomite with grain size approximately 0.008 mm with lenticles of coarser (secondary) carbonate, grain size averaging 0.08 mm, scattered detrital quartz grains to 0.1 mm and scattered pyrite cubes (average 0.016 mm); also traces of zircon are present.

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Shadowy Collenia growth-lines are seen in the dolomite indicated by alternating laminae of carbonate of different grain size.

Modal mineral composition as volume percent :-	
dolomite (fine)	95
dolomite (coarse, secondary)	4
quartz	<1
pyrite	<1
zircon	<<1

A 4240 (D.H. 3-6-2D, 14.3m)

Almost pure dolomite but as dolomitic ooliths, average 0.16 mm diameter to 2 mm's ovoid or circular cells showing well marked walls of finer carbonate crystals. Cherty quartz has replaced the interior carbonate of some grains leaving the carbonate cell-wall intact; scattered fluid inclusions (approximately 1μ) are seen in carbonate. Quartz grains vary between 0.20 up to 1.3 mm in size. Stylolites marked by their peculiar undulating seams lined by opaque matter, are seen.

Pettijohn (Ref. 5) suggests that oolite structures of this type may be algal pisolites. In any case whether of organic or inorganic origin they are a shallow water facies.

Modal mineral composition as volume percent :-	
dolomite	99
quartz	1
opaque matter	<1

A 4243 (D.H. 3-6-3D, 23.5m)

Collenia felt-like structures indicated by difference in grain size of dolomite grains (.002 mm to .01 mm). Baryte showing 90° cleavages, low birefringence, occurs as short prisms located in vugs in the dolomite

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groundmass or between laminae of structures. The baryte is generally surrounded by coarse, clear, recrystallised carbonate.

Occasional detrital quartz and fine scattered opaques (approximately 0.01 mm) are present.

Modal mineral composition as volume percent :-

dolomite	99
baryte	1
quartz	<1
opaque minerals	<<1

A 4244 (D.H. 3-6-3D, 34.0m)

Pale dolomite, showing undulating and bifurcating Collenia structures. As in previous sections, the banding is indicated by differences in dolomite crystals size (0.006 to 0.02 mm). Scattered fluid inclusions (approximately 1μ) are seen in clear dolomite grains.

Grains of detrital quartz (approximately 0.06 mm), irregular or sharply angular, occur scattered through the section. Traces of baryte (approximately 0.06 mm) occur where bands of recrystallised dolomite are seen forming part of Collenia laminae.

A stylolite traversing the section interrupts and displaces the Collenia structure and a residue of baryte with some recrystallised carbonate forms along the stylolite. A thin, even vein of recrystallised carbonate (0.2 mm thick) traverses the Collenia structure without displacement and joins the stylolite at an oblique angle.

Modal mineral composition, as volume percent :-

dolomite	99
quartz	1
baryte	<1

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A 4248 (D.H. 4-6-4D, 22.2m)

Wavy, lenticular Collenia forms are seen faintly, the laminae showing up as differences in grain size from average 0.01 to 0.024 mm. Ovoid structures as well as continuous almost parallel undulating laminae are observed. The coarser carbonate show the zones between the actual narrower Collenia felts. Laths (approximately .2 x .3 mm) of baryte intergrown in sheaves and baryte as granules with characteristic rectangular cleavage, straight extinction and with low birefringence are seen in vugs in the dolomite. A little cherty quartz occurs scattered around the section associated with coarse recrystallised dolomite in vugs.

Modal mineral composition as volume percent :-

dolomite	99
baryte	1
detrital quartz	<1
chert.	<1

In the Woocalla Dolomite group, sulphide is seen only in the basal dolomite 'shale' as fine specks. The dolomite from Woocalla Quarry contains wad and this could have soaked up any copper formerly present in sulphides and now pseudomorphed by clots of limonite in the rock.

The Mystery/Pernatty Lagoon dolomites show Collenia growths in abundance and the dolomite (by thin-section) is seen to be almost pure. Collenia is known to have grown in muddy waters on inter-tidal mud flats (Ref. 3) and in these specimens there is a general dearth of particulate sand, silt and clay-size sediment apart from dolomite. These Collenia therefore grew on dolomite-mud banks around or within a lagoon which contained appreciable saline water, rather different from the present-day condition of the mainly dry Pernatty Lagoon around the shores of

which the fossil Collenia reefs are situated. The catchment around the Lagoon was low-lying so that little or no clastic material was washed in by flood waters. Evaporation of carbonate waters under saline conditions would produce a dolomite mud almost free from clastic impurities. But whence the copper?

4.9 Whyalla Sandstone/Main Open Cut

All sections A 4251, 4253 and 4255; and A 4011 collected in July 1971 are texturally different except in rounding of the larger quartz grains.

A 4251 (D.H. 3-4-1D, 6.4m) - Rounded, Equigranular

Rounded to subangular quartz grains and more equigranular than in other Whyalla Sandstone samples collected. However, diagenetic activity has expelled most of the clay matrix and growth on originally spherical quartz grains (seen by relict particulate hematite trains and few fluid inclusions - to 2μ) has produced ragged margins around many grains and a quartz mosaic of complex sutured grains has resulted, varying between 0.1 and 0.7 mm, average 0.6 mm. Quartz rock fragments, rounded to angular, made up of tiny quartz grains (average 0.06 mm) as mosaics, are common and also some quartz gneiss grains. Rounded, brown-red hematite mottled chert grains are fairly common (average 0.4 mm). Traces of kaolinite occur in chert grains and in interstices.

Modal mineral analysis as volume percent :-

quartz mineral grains	92
quartz rock fragments	5
chert grains	3
sericite/kaolinite	< 1
hematite (in chert)	<< 1
opaque minerals	< 1

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A 4253 (D.H. 3-4-2D, 7.5m) - poorly sorted, inequigranular

A very poorly-sorted sandstone containing large clasts of angular quartz rock (to 3 mm). Outside the grossly ill-sorted zone is a layer where smaller subangular grains (approximately 0.3 mm) occur with occasional large rounded grains (approximately 1 mm) to be seen. The subangularity of the small grains is seen locally to be due to extensive silicification of the grains. Curved boundaries between quartz grains is not common but where it occurs it shows unstable pressure solution conditions during diagenesis. Many grains show strain shadow effects apparently unrelated to the present fabric since they occur scattered through the rock fabric among grains which show no strain.

Quartz rock fragments are sparse and occur in the larger size ranges to 1 mm. Chert grains are sporadic, some are chalcedonic and are approximately the size of average quartz grains overall (0.42 mm). Quartz mineral grains vary from 0.1 to 0.8 mm. Chert matrix has replaced much of the original kaolinite except in most cases for a narrow selvage of clay between the quartz grains and central core of chert in channelways. Hematite dust present is largely within chert grains or is residual in chert matrix. Cavities in the rock surrounded by narrow zones of chalcedonic fibres, scattered zircon grains and a little rutile are seen in the interstices associated with opaque detritus.

In areas where both complex suturing and straight-line contacts occur between quartz grains it appears that the latter type of diagenetic crystallisation occurred before the former and overprints it.

Modal mineral comparison as volume percent :-

quartz mineral grains	93
quartz rock fragments (mainly gneiss)	2
chert grains (mainly chalcedony	3
chert matrix	2
kaolinite, hematite, rutile, zircon, opaques	<1

A 4255 (D.H. 3-4-3D, 6.1m) - fairly equigranular and much altered by silicification

Similar to A 4253 in state of recrystallisation but better sized. Angularity of originally rounded grains is very marked in this specimen; complex suturing of mutual, parallel, interpenetrating finger-like growths between grains. Many grains appear to have completely recrystallised and pockets of chalcedony occur where possibly kaolinite matrix previously resided. Hair-line fractures have developed in some quartz grains. It was noted that in mineralised specimens these are frequently filled with atacamite. Quartz mineral grains 0.1 to 1 mm, average 0.5 mm; similar size range for quartz rock fragments. Chert grains 0.1 to 0.5 mm average 0.4 mm. Interstices are studded with occasional rounded zircon (approximately 0.08 mm) and prismatic, stumpy tourmaline grains.

Modal mineral composition as volume percent :-

quartz mineral grains	91
quartz rock fragments (some gneiss)	5
chert grains	3
chert matrix	1
hematite particles, zircon, tourmaline	<1

A 4011 (Collected in July 1971 : See Specimen 9A - Ref. 2)

Very similar to A 4253 in being poorly sorted and of similar fine and coarse grain sizes, but diagenesis (silicification) here is incomplete with up to 3 volume percent kaolinite remaining in channels of only a few

microns width between quartz grains. The quartz rock to mineral ratio is rather higher for A 4011 in being 91:5 as against 92:2 for A 4253.

There is also a greater abundance of chert fragments and matrix in A 4253.

4.10 West Lagoon (see Pacminex Report PMR. 270/71 - Ref. 2)

A 4001, 4003, 4004, 4007 and 4008 (Collected in July 1971)

There is generally more chert in the matrix than was reported in PMR. 270/72 but still appreciable kaolinite present. Specimens A 4003, 4004, 4007 and 4008 are all very similar in texture with large spherical quartz mineral grains mainly, but also some very round rocks fragments and chert grains varying between 0.05 and 1.6 mm but there is a 'size-range' gap between 0.05 and 0.2 mm. The small angular and subangular quartz grains occupying the interstices between large grains constitute a separate silt generation of different origin from the much larger round grains of aeolian origin. The matrix, apart from silt sized quartz grains is chert replacing kaolinite or sulphide. In specimens 4004, 4007 and 4008 there is an abundance of rounded chert grains.

Specimen A 4001 is very different from any of the other West Lagoon, Whyalla Sandstone specimens collected in that it contains moderately well sorted, rounded to subangular quartz grains of 0.2 to 0.7 mm and having a fair amount of kaolinite cement remaining between the grains in patchy areas. Otherwise quartz grains are packed tightly together and are fused together by secondary quartz to give a subangular growth shape to previously rounded grains indicated by narrow traces of fine hematite particles. This specimen resembles much more, specimens of Whyalla Sandstone described from the Main Open Cut that are more or less equigranular such as A 4251 and A 4255.

Specimens A 4003, 4004, 4007 and 4008 are interesting in that they are mineralised and also contain two generations of different provenance :-

- one from a suspected aeolian source and another containing the mineralisation with clay minerals and silt sized particles and thought to have been dumped into shallow channels together by flash flood waters working over playa lakes and sand dunes adjoining the lakes.

4.11 Mystery/Pernatty Lagoon

A 4242 (D.H. 3-6-3D, 17.6m) - fairly equigranular; some interstitial carbonate

Mainly equigranular without the large rounded grains characteristic of most of the Whyalla Sandstone in the district. Somewhat similar in grain shapes and sizes to A 4001 of the Whyalla Sandstone of West Lagoon and does not appear to occur essentially at any particular horizon of the stratigraphic unit but is just a local horizon established when sedimentation was less dynamic.

The quartz grains vary from 0.05 to 0.48 mm, average 0.24 mm, they range from rounded to subangular with a moderate matrix of carbonate and less kaolinite/sericite. The quartz grains are largely welded together by quartz where carbonate is absent, showing an interstitial slightly curved or straight sliver of secondary quartz a few microns in width. Occasionally the secondary quartz growth seen beyond the hematite particle trains and fluid inclusions flanking primary quartz shows optical continuity of grain growth but frequently there is displacement of the growth pattern shown by a different optical orientation.

Quartz rock fragments, average 0.24 mm, are fairly common and are varied - gneiss, fine mosaic, etc.

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Scattered, rounded chert grains of similar size are speckled with opaque hematite/limonite.

Microcline feldspar shows incipient sericitisation along cleavages but still displays twinning; albite occurs as clear square prisms with sharp albite twinning and is possibly diagenetic. Opaque minerals occur as large anheda, average size 1.0 mm. Carbonate occurs as cement between quartz grains or as clear interstitial anheda, average 0.10 mm.

Modal volume percent of minerals present :-

quartz mineral grains	89
quartz rock fragments	5
chert fragments	2
albite and microcline feldspar	1
carbonate	3
kaolinite/sericite	<1
tourmaline	<<1

A 4247 (D.H. 3-6-4D, 15.2 m) - large rounded quartz mineral grains and rock fragments as per normal Whyalla Sandstone, but much carbonate matrix

A rock with wide differences in grain size and matrix content. The large round quartz grains vary from 0.32 to 2 mm (average 0.32 mm). The 2 mm diameter grain is a rock fragment. One large rounded albite grain, 0.5 mm diameter altering to kaolinite along cleavages is seen. Smaller, mainly angular to subangular quartz grains or roughly 0.080 mm equigranular size are just above the maximum for silt size (0.062 mm) and are of different provenance from the larger grains. They are accompanied by similar sized grains of chert, kaolinite and a little albite.

Large rounded chert grains up to 2 mm of similar size ranges to the quartz grains occur, some show a little

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kaolinite/sericite development as seen in A 4242 as well as of fine opaque matter.

Apart from being present as fine matrix the carbonate occurs as large angular clasts of pure carbonate to over 2 mm in size and containing ovoid algal structures. These are obviously fragments of now eroded parts of the underlying Woocalla Dolomite.

Opaque minerals as subhedra (average 0.080 mm) occur in the matrix occasionally as cement between quartz or in rock fragments.

Modal volume percent :-

quartz mineral grains (large)	68
quartz mineral grains (small)	10
quartz rock fragments	2
chert fragments	2
albite feldspar	< 1
carbonate matrix	7
carbonate clasts	10
kaolinite/sericite	< 1
opaque minerals	1

The Whyalla Sandstone contains many large rounded grains of aeolian origin, mainly quartz mineral grains but the largest are also of quartz rock fragments and chert from some previous eroded rock formation. The matrix quartz grains are frequently quite angular and as previously suggested are silt from one time neighbouring playa lakes.

Variation in composition of clasts and matrix may reflect the nature of adjoining and/or underlying deposit as for example Woocalla Dolomite beneath the Whyalla Sandstone in the Pernatty Lagoon/Mystery area; Pandurra Quartzite beneath the Whyalla Sandstone in the Main Open Pit. A periodic slowing-up in the speed of

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deposition of the Whyalla Sandstone produces the equigranular fine to medium grained subangular quartz grain deposits.

Constant recycling of clastic material in the area has eliminated all but a minimum of detrital heavy minerals generally of minimal zircon or tourmaline.

The presence of fresh microcline indicates a dry palao climate with periodic violent flooding. Now see below.

4.12 Plateau Exposure (just north of the Main Open Pit)

A 4234

In hand specimen feldspars appear of roughly similar size to rounded quartz grains but some have oblong shape and alteration to clay mineral is observed.

The large round quartz grains (0.3 to 1.3 mm, average 0.6 mm) have grown by quartz addition from percolating solutions containing silica to fuse with neighbouring grains. Here the original grain limits are rarely demonstrated by hematite particles but more frequently by a fine ovoid or circular 'crack' containing fine opaque particles.

Quartz rock fragments are rounded and of similar grain size to rounded quartz mineral grains.

Large rounded chert grains (approximately 1.2 mm), show outgrowths of tiny quartz in mosaics to join continuous overgrowths of neighbouring quartz grains.

The feldspar grains of a polyhedral or rounded outline are decomposed to kaolinite, range up to 1.2 mm, average 0.5 mm.

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The absence of second generation of fine quartz or chert fragments is noted.

Modal volume percent :-

quartz mineral grain	93
quartz rock fragments	4
microcline feldspar (now kaolinite)	2
chert grains	1

No heavy mineral grains are present.

A 4235

Very similar to A 4234 except that quartz rock fragments are more variable in the larger size ranges, though the average is about the same and quartz grains contain occasional flakes of muscovite (approximately 0.020 mm).

Modal mineral volume percent :-

quartz mineral grains	92
quartz rock fragments	5
chert grains	1
microcline feldspar (now kaolinite)	2
tourmaline	< 1
opaque mineral	< 1
muscovite	<< 1

4.13 Tregolana Shale : Pernatty Lagoon/Mystery Area

A 4239 (D.H. 3-6-2D, 5.0m)

A fine grained intergrowth of cherty quartz (maximum 0.050 mm, average 0.015 mm), sericite flakes (average 0.04 x 0.015 mm) and fine opaques and hematite particles (range 0.003 to 0.050 mm, average 0.006 mm) in the interstices. Scattered carbonate granules (0.013 to 0.026 mm) occur.

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Fine, parallel, linear concentrations of quartz grains, sericite and hematite particles gives a banded appearance to the rock.

The quartz is clear of inclusions and variable in shape from angular to rounded, and is similar to the fine matrix quartz - in its varied shapes-in most Whyalla Sandstone samples.

Modal mineral composition as volume percent :-

quartz	78
sericite	17
carbonate	1
hematite	3
opaque minerals	1

A 4246 (D.H. 3-6-4D, 5.5m)

Similar to A 4239 in nature of minerals, sizes, shapes and banding but the proportions of sericite and quartz are different and with the additional presence of kaolinite this rock is more argillaceous.

Quartz grains, range 0.016 to 0.11 mm, average 0.017 mm.

Sericite flakes, range 0.03 to 0.10 mm, width approximately 0.03 mm.

Hematite/opaque grains 0.004 to 0.02 mm, average 0.006 mm.

Modal mineral composition, volume percent :-

quartz	20
sericite	72
kaolinite	5
carbonate	1
hematite/opagues	2

Variations in mineral concentrations shown by banding indicate seasonal variations in climate, i.e. velocity of run-off water entering the depositional environment, since apart from fine bands or lenticles of carbonate and kaolinite rich bands, the nature of minerals present as shown by drill core is fairly constant with depth in this stratigraphic unit in the Mystery area.

4.14 Mount Gunson : Corruburra Sandstone

Some variations in appearance were noted in the field and hand specimens of each of the varieties seen were collected and sectioned.

A 4222 (chocolate brown, coarse grained)

Grain shapes of quartz are slab-like and angular to ovoid; many grains were originally ovoid as indicated by interal fine particle limonite trains but have been made angular by diagenetic silica growth; now fairly equigranular. Many grains show strain shadows and some lath-like grains show Böhm lamellae and are apparently of metamorphic origin. Many grains are completely strain free. Quartz grains range from 0.05 to 0.24 mm, average 0.09 mm. Hematite/limonite particles form a cement where quartz growth due to diagenesis has not penetrated the interstices. The limonite cement is of fairly regular thickness (to 0.009 mm) and limonite also occurs densely in sporadic chert grains which themselves are similar in shape and size to quartz grains. Quartz rock fragments are also of similar size to quartz mineral grains and are of minor importance.

Microcline as fresh grains are detrital, subangular, and of similar grain size to neighbouring quartz grains (to 0.2 mm, average 0.08 mm) and cemented with hematite/limonite.

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Heavy minerals are shown by scattered rounded barrel-shaped zircon grains (approximately 0.08 mm) clear of debris and yellowish-green detrital pleochroic tourmaline of similar grain size.

Clay mineral as kaolinite platelets occupies scattered pockets in the rock of up to 0.3 x 0.15 mm in size.

Muscovite, detrital, as long thin plates (to 0.3 mm) are present in trace amounts.

The tabular, quartz-muscovite, micaceous chert and some clay pockets give a very rudimentary bedding to the rock.

Modal mineral volume percent composition :-

quartz mineral grains	86
quartz rock fragments	1
microcline	< 1
chert grains	1
micaceous chert	3
muscovite	1
limonite pseudomorphs of magnetite (?)	5
limonite/hematite, interstitial	2
zircon	1
tourmaline	< 1

A fairly well-sorted sandstone which contains rather more heavy minerals but of still limited quartz grain size variation compared with the underlying Whyalla Sandstone. A moderate abundance of lamellar strained quartz shows a metamorphic source for much of this quartz.

A 4223 (red-brown and flaggy)

More coarse grained than A 4222, more equigranular but marked by finer and coarser bands of quartz. This sample is cleaner, with less cementing matrix of hematite/limonite. The grains are rounded to subangular. Range of grain sizes in different bands 0.1 to 0.8 mm, average 0.3 mm; 0.05 to 0.4 mm, average 0.2 mm.

As in the previous section described, the quartz is present mainly as mineral grains. Few grains show strain-shadows, otherwise the grains are relatively strain-free. Similarly, many grains are joined one to another through diagenetic addition of quartz to the grains but many have retained a hematite/limonite cement of maximum 0.008 mm thickness.

The usual fine trains of hematite particles within quartz grain margins indicate a slightly more rounded grain before diagenesis, although in this rock silica diagenesis was very slight and the hematite particles are just within the quartz grain boundaries. Close packing of quartz grains before diagenesis occurred leaving minimal space for quartz growth, although isolated pockets of clay occur measuring from 0.06 to 0.65 mm square and some pockets contain a high proportion of limonite.

A few large (approximately 0.1 mm) detrital zircon grains are seen but traces only of opaque mineral are present which are now pseudomorphed by limonite.

No feldspar is seen.

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Modal mineral volume percent composition :-

quartz mineral grains	92
quartz rock fragments	1
kaolinite	5
hematite/limonite	2
opaque minerals	< 1
zircon	< 1

A 4224 (pale, leached, massive sandstone)

Similar in grain size to A 4222 but less matrix and abundant magnetite. The quartz grains are subangular to rounded, range 0.03 to 0.50 mm, average 0.10 mm. Few quartz rock fragments of similar size are seen. However many quartz grains show strain shadows, Böhm lamellae and other evidence of strain. However, since they are scattered among many grains which exhibit no strain effects it is assumed that many of them are derived from metamorphic rocks of distant provenance. Many quartz grains show fused, faintly curved boundaries and many show an inner more rounded grain outlined by hematite particles or fluid inclusions ($< 1\mu$) adjoining grain boundaries.

Scattered oblong fresh microcline grains, average 0.1 mm, are present, also muscovite flakes as fine shreds to 0.3 x 0.1 mm, but commonly 0.1 x 0.04 mm and with frequent elongated yellowish clay (illite ?) flakes in pockets give a banded appearance to the rock.

A little clear wedge-shaped sphene, rounded zircon (0.08 mm) and tourmaline as detrital grains are seen.

Magnetite (or limonite pseudomorphs), interstitial, around 0.05 mm.

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Some concentration of heavy minerals has taken place here yet the even-grained nature of quartz does not suggest turbulent current action.

Modal mineral volume percent :-

quartz mineral grains	83
quartz rock fragments	1
microcline	1
kaolinite/illite ?	8
magnetite (mainly altered to limonite)	6
sphene, zircon, tourmaline	1

4.15 Mount Gunson : Simmens Quartzite

A 4221

In hand specimens it is seen to be markedly current bedded.

The thin-section shows an approximately even-grained mosaic of quartz and virtual absence of soft clay, limonite, etc, matrix between the quartz grains. Coarser grained than Corruburra Sandstone (A 4222, 4224) and more even grained than A 4223. Fairly equigranular at range 0.08 to 0.32 mm, average 0.24 mm. Diagenesis has produced quartz grains of subangular aspect with gently curved contacts, showing an originally more rounded primary nucleus, enclosed by fine hematite particle trains in some cases. Flattened, angular or ovoid chert grains of roughly similar grain size (average 0.16 mm) to quartz mineral grains occur. The chert grains vary somewhat in matrix composition with apart from minute quartz grains (0.010 to 0.015 mm size) they also contain fine flakes of mica (clay ?) and/or hematite flakes.

Some chertification is seen around edges of occasional quartz grains.

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Fluid inclusions (up to 5μ) occur abundantly within some quartz grains but are not abundant along quartz grain boundaries.

Fairly abundant fresh microcline as stumpy prisms and some kaolinite pseudomorphs of microcline, range 0.08 to 0.24 mm, average size 0.16 mm are seen scattered through the section.

Trace amounts of tourmaline, zircon, topaz and purplish fluorite are present.

The rock could be called a sub-feldspathic lithic arenite - Williams, et al (Ref. 7).

Modal mineral volume percent constituents :-	
quartz mineral grains	92
chert grains	2
microcline	4
zircon, tourmaline, sphene, purplish rutile	~ 1
apatite ?	1

Many quartz grains show strain shadows but only very few show lamellae strain characteristics.

4.16 Fluid Inclusions

Quartz mineral grains and rock fragments were examined in detail for fluid inclusions.

The quartz grains of all rock units contained fluid inclusions, usually between 1 and 2 microns in size, sometimes to 5 microns. The smaller inclusions are ovoid, round or oblong; larger inclusions are generally somewhat irregular and have the appearance of sealed-off capillaries.

Both Whyalla Sandstone and Pandurra Quartzite contain quartz grains with abundant fluid inclusions. These are mainly in narrow or broad zones of overgrowths

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around primary quartz grains, frequently the zone begins approximately 5 microns in from the wall of the grain. Fluid inclusions adjoining the wall of quartz grains are frequently intermixed with tiny hematite platelets. Planes of weakness due to stress will often contain myriads of fluid inclusions, occasionally as small as 0.1 microns, but are also large (to 5 microns) and are frequently strung out as continuous lines of tubular inclusions which originally together formed capillaries that have become divided by quartz deposition. Gneiss grains which have recrystallised, show streams of inclusion along the constituent mineral grain boundaries and chert grains also contain large numbers of tiny disoriented fluid inclusions.

Commonly, tiny gas bubbles are seen in the smaller inclusions, approximately 2 microns in size and also occasionally in the larger inclusions. Each gas bubble is of fairly constant (5 to 10 volume percent) of total enclosing cavity but occasionally is present in higher ratio to liquid. Fewer grains show crystal inclusions particularly if a gas bubble is present. In one West Lagoon, Pandurra Quartzite specimen a tiny cube crystal of halite was seen. In the Main Open Pit Pandurra Quartzite specimens, tiny flakes of hematite (?) are seen in the fluid of the inclusions.

Fluid inclusions are particularly abundant in the Main Open Pit Whyalla Sandstone and Pandurra Quartzite specimens, scattered through quartz overgrowths, or along planes of weakness in the quartz grains. Scattered fluid inclusions are seen in malachite in the Pandurra Quartzite but no gas bubbles are seen.

In West Lagoon specimens the inclusions to a great extent follow outer zones of quartz overgrowth formed during diagenesis.

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Fluid inclusions in the Pernatty Lagoon/
Mystery specimens of Woocalla Dolomite are sparse and tend to be confined to the boundaries of small carbonate grains. Very few are embedded in individual grains. In the Pandurra Quartzite, fluid inclusions are as abundant and are distributed as per the Pandurra Quartzite of West Lagoon.

The Pandurra Quartzite of Woocalla Quarry show abundant fluid inclusions in overgrowths associated with particles of hematite and limonite, also in diagonal cross zones of weakness in recrystallised quartz gneiss grains as fluid inclusions only. Very few gas bubbles are seen in fluid inclusions here.

In the Whyalla Sandstone from the "Plateau area" west of the Mill, fluid inclusions occur patchily inside quartz grains to 5 microns and are of rather irregular size; some bubbles occur in fluid. Zonal distribution in overgrowths is thin or absent.

Simmens Quartzite, quartz grains show very fine inclusions in criss-crossing diagonal planes (zones of weakness), some contain gas. Few fluid inclusions are seen to occur.

The Cornuburra Sandstone grains show some planes of weakness where tabular inclusions (approximately 4 x 2 microns) with long axes parallel to the plane indicate original fine fissures which trapped fluid and then became sealed by secondary quartz.

As a matter of interest zircon, tourmaline and on an occasion apatite grains are seen embedded in quartz overgrowths alongside fluid inclusions.

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